SUPPLEMENT TO: THE GENERALIZED FERMAT EQUATION $x^3 + y^3 = z^n$

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This Supplement belongs to the paper "The generalized Fermat equation $x^3 + y^3 = z^n$". Its items are numbered S1, S2, ...; all other numbers (of sections, statements, displays and of the items C1–C5 of Appendix A) refer to the paper. The Supplement contains the computations for Part IV of the paper, computations that check results proved in the text of the paper, explicit Heisenberg extensions for the five planes used in the proof of Proposition 11.10, a second proof of a vanishing in Part I, and numerical evidence at the point of the Catalan solution, with the details of the numerical check of Remark 4.18. The description of the computations at the beginning of Appendix A applies to the items of this Supplement other than the numerical checks of the explicit formula in item S5 and the numerical fits of item S25, which are reported as evidence only; the programs, their outputs and the commands that run them, including those of these fits, are in [1].

Part I

S1. The vanishing outside degree two. This is a second proof of part of Proposition 3.3 (a), with the notation of Proposition 3.4. The vanishing $H_c^i(X_s)^{\chi^m} = 0$ for $i \neq 2$ also follows directly: X_s is a smooth affine surface, so $H_c^0 = H_c^1 = 0$ by Artin vanishing and duality; H acts trivially on H_c^4; and $H_c^3(X_s)^{\chi^m} \cong H^3(\tilde{X}_s)^{\chi^m}$ is dual to $H^1(\tilde{X}_s)^{\chi^{-m}} = H_c^1(X_s)^{\chi^{-m}} = 0$.

S2. Further checks of Part I. The record [1] contains numerical checks of Lemma 6.1, Proposition 5.1 and Lemma 5.2.

Part II

S3. Finite groups. This is a check of Lemma 8.5. A computation with GAP [4] confirms that the non-abelian simple groups of order less than 3600 and the orders of their automorphism groups are those listed in its proof.

S4. Class fields. This is a check of Lemma 8.6. The Minkowski bounds of K, $K(\sqrt{5})$ and $\mathbb{Q}(\zeta_{15})$ are $1.1026\ldots$, $2.2797\ldots$ and $7.1054\ldots$, and none of these fields has a prime ideal of norm below its bound; the ray class group of K of conductor $5\mathcal{O}$ is cyclic of order 4, and that of $K(\sqrt{5})$ of conductor the prime above 5 has order 2.

S5. Further checks of Part II. The record [1] contains numerical checks of the explicit formula (8.1) for five fields, and of Lemma 8.4 for the subgroups of $\mathrm{GL}_2(\mathbb{F}_5)$, $\mathrm{SL}_2(\mathbb{F}_{25})$, $\mathrm{SL}_3(\mathbb{F}_5)$ and $\mathrm{SU}_3(\mathbb{F}_5)$. The inequalities of Proposition 8.3 for $n \geq 18$ and $n \geq 120$ can also be read off from the numbers printed in Odlyzko's tables of 1976 [2]: 9.280 and 17.020 for totally complex fields of degree at least 18 and 120.

Part III

In the items of this section, PARI's `bnfinit` is used only for K and for the three fields $K(x^{1/3})$ of item S8, with $x = 1-\omega, 2, \omega$. The field K has class number one (item S4) and unit group μ_6. The three sextic fields, of discriminants -3^{11}, $-2^4 \cdot 3^7$ and -3^9, have class number one: their Minkowski bounds are less than 14, 6 and 5, and every prime ideal of norm below the bound is principal, with an explicit generator [1]. Their unit groups enter only through PARI's `rnfconductor` in item S8, and the breaks it gives agree with those computed from relative discriminants, which use no class group or unit group. So no statement depends on the Riemann hypothesis.

S6. Cosquares. This is a check of Lemma 10.1. For $q = 3, 5, 7, 11, 13$ every $A \in \mathrm{GL}_3(\mathbb{F}_q)$ was enumerated (up to scalars, which do not change $A\,{}^tA^{-1}$), and $M = A\,{}^tA^{-1}$ was tested for being a scalar times a unipotent matrix. In every such case $M = 1$, which happens exactly for the $q^2(q-1)(q^3-1)$ symmetric matrices A, or M is a regular unipotent matrix, which happens for 3 744, 297 600, 4 826 304, 193 116 000 and 748 185 984 matrices respectively; and $\mathrm{rank}(M-1) = \mathrm{rank}(A - {}^tA)$ always.

S7. Cubic characters of $K_\mathfrak{p}$. This is a check of Lemma 11.2. By enumeration: $(\mathcal{O}/9)^\times$ has order 54 and is generated by $1+\omega$, 4, $1+3\omega$, of orders 6, 3, 3; the cube of every principal unit modulo 81 is $\equiv 1 \pmod 9$; and $1 + (1-\omega)^3 \equiv 16(1+3\omega) \pmod 9$. The cubic Hilbert symbol $(x, y)_\mathfrak{p}$ was computed from the product formula, as the inverse of the product of the tame symbols at the other places, and tested on 400 random pairs for the Steinberg relation, $(x, -x) = 1$, antisymmetry, and agreement with the bilinear form of Lemma 11.2 (c) on the coordinates. Of the 130 planes of $\mathcal{H}$, exactly 40 are isotropic, and 13 of these meet $\mathcal{U}$; the same 40 planes were found independently as those $\langle \alpha, \beta \rangle$ for which a global representative of β is a norm from $K(x^{1/3})$, x a global representative of α. The upper breaks of the cubic extensions cut out by the 40 lines of $\mathcal{H}$, computed from PARI's relative discriminants of global models $K(x^{1/3})/K$, agree with Lemma 11.2 (b). As a check of Proposition 11.1: the ray class groups $\mathrm{Cl}_{\mathfrak{p}^n}(K)$ have order $3^{\max(n-2,0)}$ for $n \le 20$ (PARI's `bnrinit`, and for $n \le 12$ an enumeration of $(\mathcal{O}/\mathfrak{p}^n)^\times$ that does not use PARI).

S8. Worked Heisenberg extensions. We use the following converse of Lemma 11.5. Let $K_1/K_\mathfrak{p}$ be cyclic of degree 3, cut out by $\alpha \in \mathcal{H}$, with generator σ. Let $w \in K_1^\times$ be such that the class of $\mathrm{N}(w)$ in $\mathcal{H}$ is not in $\mathbb{F}_3\alpha$, let $f \in K_\mathfrak{p}^\times$, and put $z = \mathrm{N}(w)$ and $\gamma = \sigma(w)^2\sigma^2(w)$. Then $K_1(z^{1/3}, (f\gamma)^{1/3})$ is a Heisenberg extension of $K_\mathfrak{p}$ [3, Theorem 2.5] (the computation of s^3 in the proof of Lemma 11.5 also shows this), and changing f changes c in Corollary 11.7 by an element of $\mathcal{H}$.

Let $x \in K_\mathfrak{p}^\times$ not a cube, θ a cube root of x, $K_1 = K_\mathfrak{p}(\theta)$, $\sigma(\theta) = \omega\theta$, and $w = 1 + b\theta^j$ with $b \in K_\mathfrak{p}$ and $j \in \{1, 2\}$. Then $z = \mathrm{N}_{K_1/K_\mathfrak{p}}(w) = 1 + b^3x^j$. If the class $[z]$ is not in $\mathbb{F}_3[x]$, the classes $[x]$, $[z]$ span an isotropic plane W by the Steinberg relation, and with $\gamma = \sigma(w)^2\sigma^2(w)$ the field $L = K_1(z^{1/3}, \gamma^{1/3})$ is a Heisenberg extension of $K_\mathfrak{p}$ with plane W (by the converse above). The same x and w over K give a global field L' of degree 54 over $\mathbb{Q}$ whose completion at $\mathfrak{p}$ is L, and $v_3(d_{L'}) = 27(1+d)$, where d is the valuation of the different

of $L/K_\mathfrak{p}$ normalized as in the proof of Proposition 11.11. The following five planes are used in the proof of Proposition 11.10.

(1) $W_1=\langle e_\pi,e_2\rangle$, of type (i): $x=1-\omega$, $w=1-\theta$, $z=\omega$. Here $t=3$, θ is a uniformizer, and $1-\omega=\theta^3$, so
$$\frac{\sigma(w)}{w}=1+\frac{(1-\omega)\theta}{1-\theta}=1+\frac{\theta^4}{1-\theta}\in U_{K_1}^{(4)}\smallsetminus U_{K_1}^{(5)},$$
of break $9-4=5$ over K_1 (Lemma 11.3 (c)), while $z=\omega=(1-\theta)^3(1+3\theta/(1-\theta)^2)$ has break 2 over K_1. The largest upper break is $\varphi_{K_1/K_\mathfrak{p}}(5)=11/3$; the upper ramification groups have orders 27 for $y\le 2$, 9 for $2<y\le 3$, 3 for $3<y\le 11/3$, so $d=3\cdot\frac{26}{27}+\frac{8}{9}+\frac{2}{3}\cdot\frac{2}{3}=\frac{38}{9}$ and $v_3(d_{L'})=141$.

(2) $W_2=\langle e_\pi,e_1+e_0\rangle$, of type (ii): $x=1-\omega$, $w=1+\theta^2$, $z=1+x^2=1-3\omega$, of class $-(e_1+e_0)$. Here $t=3$, $\sigma(w)/w=1+(\omega^2-1)\theta^2/(1+\theta^2)\in U_{K_1}^{(5)}\smallsetminus U_{K_1}^{(6)}$ has break 4, and z has break at most $\psi_{K_1/K_\mathfrak{p}}(1)=1$ over K_1; the largest upper break is $\varphi_{K_1/K_\mathfrak{p}}(4)=10/3$, and $v_3(d_{L'})=133$.

(3) $W_3=\langle e_\pi+e_2,e_1\rangle$, of type (ii): $x=2$ (of class $-e_1$, break 1), $\theta=2^{1/3}$ a unit of K_1, $w=1+\theta$, $z=3$, of class $-(e_\pi+e_2)$. As $v_{K_1}(1+\theta)=2$ and $v_{K_1}(\omega-1)=3$, $\sigma(w)/w=1+(\omega-1)\theta/(1+\theta)\in U_{K_1}^{(1)}\smallsetminus U_{K_1}^{(2)}$ has break 8, and z has break at most $\psi_{K_1/K_\mathfrak{p}}(3)=7$; the largest upper break is $\varphi_{K_1/K_\mathfrak{p}}(8)=10/3$, and $v_3(d_{L'})=133$. This is the field $\mathbb{Q}(\omega,2^{1/3},3^{1/3},\gamma^{1/3})$ with $\gamma=(1+\omega 2^{1/3})^2(1+\omega^2 2^{1/3})$.

(4) $W_4=\langle e_2,e_0\rangle$, of type (iii), with $W\cap\mathcal{U}=\mathbb{F}_3 e_0$: $x=\omega$, $\theta=\zeta_9$ ($t=2$), $w=1+(1-\omega)\zeta_9$, $z=7+3\omega$, of class e_0. Here $\sigma(w)/w\in U_{K_1}^{(6)}$ has break at most 2 (Lemma 11.3 (b)) and z is unramified over K_1, so the largest upper break is at most $\varphi_{K_1/K_\mathfrak{p}}(2)=2$, hence equal to 2, the break of e_2; and $v_3(d_{L'})=99$.

(5) $W_5=\mathcal{U}=\langle e_1,e_0\rangle$, of type (iii): $x=2$ ($t=1$), $w=1-(1-\omega)\theta$, $z=7+12\omega$, of class e_0. Here $\sigma(w)/w\in U_{K_1}^{(6)}$ has break at most 2, and z is unramified over K_1, so the largest upper break is at most $\varphi_{K_1/K_\mathfrak{p}}(2)=4/3$; it is 1, and $v_3(d_{L'})=75$.

In (4) and (5) the values of the largest upper break and of $v_3(d_{L'})$ follow from Remark 11.9. Replacing γ by $f\gamma$ changes c by an element of $\mathcal{H}$, and in case (iii) it can change the largest upper break: in (4), $f=1-\omega$ gives the breaks $-,5,5,5$ over K_1 (the line of z unramified), largest upper break 3 and $v_3(d_{L'})=117$; in (5), $f=\omega$ and $f=1-\omega$ give the breaks $-,4,4,4$ and $-,7,7,7$, largest upper breaks 2 and 3, and $v_3(d_{L'})=93$ and 111. For each of the eight models, $v_3(d_{L'})$ was also computed by method (i) of item S9, and $v_3(d_{L'})-27$ in three further ways, which agree: from the filtration, as $27d$; by the tower formula $\sum_\delta v_{K_1}(d_{K_1(\delta^{1/3})/K_1})+9\,v_\mathfrak{p}(d_{K_1/K_\mathfrak{p}})$ over the four lines δ of the Kummer group ($K_1/K_\mathfrak{p}$ is totally ramified in all eight models); and by the conductor–discriminant formula.

S9. The Heisenberg extensions. This is a check of Proposition 11.6. For an isotropic plane W, a global model is constructed as follows: $x\in K^\times$ represents a ramified line $\langle\alpha\rangle\subset W$; $K_1=K(x^{1/3})$, with $\sigma\in\mathrm{Gal}(K_1/K)$ a generator; $w\in\mathcal{O}_{K_1}$ is chosen so that $y=\mathrm{N}_{K_1/K}(w)$ represents a class of $W\smallsetminus\langle\alpha\rangle$; and $L=K_1(y^{1/3},\gamma^{1/3})$ with $\gamma=\sigma(w)^2\sigma^2(w)$. Then L/K is Galois with group the Heisenberg group of order 27, the decomposition group at $\mathfrak{p}$ is the whole group, and the completion of L at $\mathfrak{p}$ is an extension L_W as in Corollary 11.7, for some c. Two methods were used. (i) For one model of each of the 40 planes, $v_3(d_L)$ of L as a field of degree 54 over $\mathbb{Q}$ was computed from a 3-maximal order; the breaks of the four

cubic subextensions of $K(x^{1/3}, y^{1/3})/K$ from their conductors over K; and the central break from the formula for d in the proof of Proposition 11.11. (ii) For 936 models (every ramified line α of every isotropic plane, and several w), the breaks over K_1 of the cubic characters of L/K_1 attached to y, γ, γy, γy^2 were computed by a direct valuation algorithm and by PARI's relative discriminants (they agree on all 3 744 characters), and transferred to $K_{\mathfrak{p}}$ with the Herbrand function of $K_1/K_{\mathfrak{p}}$; for 60 of these models, $v_3(d_L)$ was also computed from the relative discriminant of L/K_1 and compared with the value predicted by the ramification filtration (all agree). The two methods agree on every plane:

isotropic planes	breaks over $K_{\mathfrak{p}}$	largest break of L_W	$v_3(d_L)$
27 with $W \cap \mathcal{U} = 0$	$2, 3, 3, 3$	$11/3$	141
9 with $W \cap \mathcal{U}$ a line $\neq \mathbb{F}_3 e_0$	$1, 3, 3, 3$	$10/3$	133
3 with $W \cap \mathcal{U} = \mathbb{F}_3 e_0$, $W \neq \mathcal{U}$	$-, 2, 2, 2$	3 or 2	117 or 99
$W = \mathcal{U}$	$-, 1, 1, 1$	3, 2 or 1	111, 93 or 75

In the last two rows $-$ marks the unramified line, and the largest break, which is at most 3, depends on the choice of c; Proposition 11.6 bounds it only from above.

S10. Plücker vectors. This is a check of the linear algebra in the proof of Proposition 11.10. In the basis $(e_i \wedge e_j)$ of $\bigwedge^2 \mathcal{H}$ ordered as $(\pi 2, \pi 1, \pi 0, 21, 20, 10)$, the linear form on $\bigwedge^2 \mathcal{H}$ induced by the cup product (proof of Proposition 11.10) has coefficients $(0, 2, 1, 1, 0, 0)$ up to sign. The Plücker vectors of the 40 isotropic planes span a space of dimension 5, so they span $\mathcal{P}$, and the only nonzero linear forms vanishing on all of them are the multiples of that form; the 13 isotropic planes that meet $\mathcal{U}$ are exactly those with $\xi_{\pi 2} = 0$, and their Plücker vectors span a space of dimension 4, equal to $\mathcal{P}_0$ (which has 81 elements).

S11. Finite groups. This is a check of Propositions 11.13 and 11.14, made with the library of small groups of GAP. Among the 1 717 groups of the orders allowed by step (c) of the proof of Proposition 11.13 (3, 6; 9, 18, 36, 45; 27, 54, 108, 135, 189, 216, 270, 297; 81, 162, 324, 405, 567, 648; 243, 486), none is generated by its Sylow 3-subgroups and at the same time has no nontrivial normal 3-subgroup. The non-abelian simple groups of order at most 688, the 3-parts of their orders, and the automorphism groups of the 14 groups of order at most 12 prime to 3 are as stated there. The groups of order at most 39 with no nontrivial normal 3-subgroup that are generated by their Sylow 3-subgroups are A_4, $\mathrm{SL}_2(\mathbb{F}_3)$, $C_7 \rtimes C_3$ and $C_{13} \rtimes C_3$; among the subgroups of $\mathrm{PGL}_2(\mathbb{F}_q)$, $q = 3, 9, 27$, only A_4 occurs.

Part IV

The items of this section were computed with PARI/GP and Python. Where PARI's `bnfinit` is used, it only proposes generators: each class group is determined below the Minkowski bound by explicit generators of the prime ideals, so no statement depends on the Riemann hypothesis. The criterion of Proposition 12.11 is evaluated with the curves E_ζ and W as stated. The group $\mu_n(\mathbb{F}_q)$ is generated by g^p for a primitive root g modulo q, and a_q is computed by PARI's `ellap`.

S12. The curves $E'_{a,b}$, W and W'. This is used in the proofs of Lemma 12.7 (a), Theorem 12.9 and Theorem 12.2 (a). The identities of Lemma 12.7 (a), the translated coefficients of the proof of (d) and the model of (e) were checked as identities in $\mathbb{Q}(a, b)$. The curve $W = [0, 0, 0, 6, -7]$ has conductor 72, $j = 2048/3$ and $a_7(W) = 0$; $W' = [0, -1, 0, 1, 0]$ is isomorphic to the twist of W by -3, has conductor 24, $\Delta_{\min} = -48$, $c_4 = -32$ (so $v_2(\Delta_{\min}) = 4$, $v_2(c_4) = 5$, $v_3(\Delta_{\min}) = 1$) and $a_7(W') = 0$. The space $S_2(\Gamma_0(N))$ is zero for

every proper divisor N of 24, and $\dim S_2(\Gamma_0(24)) = 1$, the newform having the coefficients $a_q(W')$ for $5 \le q < 200$. Among the pairs (a, b) modulo 7 with $7 \nmid a^3 + b^3$, exactly 18 have $a_7(E_{a,b}) = 0$, all with $7 \nmid ab$ and $a^3 \equiv b^3 \pmod 7$.

S13. The test at the primes above 7, in product form. This is a check of item C2 of Appendix A in the form used for inert p in Proposition 12.20. For the five characters ψ of the proof of Theorem 6.2 and $\mathfrak{q} \in \{\mathfrak{l}, \bar{\mathfrak{l}}\}$, put $R_{\mathfrak{q}}(\psi) = \prod_{x \in \mu_3} P_{\mathfrak{q}}(x\psi(\mathfrak{q})) \in \mathcal{O}$. The index in $\mathcal{O}$ of the ideal $(R_{\mathfrak{l}}(\psi), R_{\bar{\mathfrak{l}}}(\psi))$ is $2^2 \cdot 3^{11}$, $2^2 \cdot 3^{11} \cdot 7^{12}$, $2^2 \cdot 3^{11} \cdot 7^{18}$, $2^2 \cdot 3^{11} \cdot 13^2$, $2^2 \cdot 3^{11} \cdot 7^2$ for the types $(0,0)$, $(1,1)$, $(2,2)$, $(0,2)$, $(2,0)$ when $(P_{\mathfrak{l}}, P_{\bar{\mathfrak{l}}}) = (P^+, P^-)$, and for (P^-, P^+) the indices of the types $(0,2)$ and $(2,0)$ are exchanged; in both cases its prime factors are (2) (the only inert one), $\mathfrak{p}$, and primes above 7 or 13.

S14. The criterion for $k = 1$ and $7 \le p < 10^8$. This is used in the proof of Theorem 12.2 (b). For each prime p the program looks for the least even n such that $q = np + 1$ is prime and (A) and (B_1) of Proposition 12.11 hold. For the 9 590 primes $5 \le p \le 10^5$ such an $n \le 112$ exists except for $p = 5$ and $p = 13$ (none with $n \le 600$); for the 654 987 primes $10^5 < p < 10^7$ it exists, with $n \le 234$; and for the 5 096 876 primes $10^7 < p < 10^8$ it exists, with $n \le 284$.

S15. The prime 13. This is used in the proof of Theorem 12.2 (b) for $p = 13$. The data given there ($a_{79}(W) = -8$, $r = 23$, $\mu_6(\mathbb{F}_{79})$, the three ζ with d_2 a square, and their X_1) were computed and checked, and the refined condition was checked to be independent of the choices of d and r.

S16. The criterion for $k \le 7$. This is used in the proof of Theorem 12.2 (c). For $k \in \{1, 3, 4, 5, 6, 7\}$ and the primes $7 \le p \le 10^6$ with $p \nmid 2k - 3$, the first form of Proposition 12.11 holds with some even $n \le 2000$, except for $p = 13$ ($k = 1$), $p = 7$ ($k = 3$), $p = 7, 13$ ($k = 6$) and $p = 23$ ($k = 7$); the largest n needed is 220. For $(k, p) = (6, 7)$ and $(6, 13)$ the refined form holds with $n = 6$ and $q = 43$ ($a_{43}(W) = 4$), respectively $q = 79$. As a control, for $k = 2$, where $2^3 + 1^3 = 9 \cdot 1^p$ is a solution, neither form holds for any prime $7 \le p \le 20\,000$ and $n \le 400$ (first form), or $7 \le p \le 2000$ and $n \equiv 0 \pmod 6$, $n \le 600$ (refined form, 7 599 pairs (p, q)).

For $k = 2$ condition (B_2) never holds: $\zeta = 1/9$ lies in the coset $9^{-1}\mu_n(\mathbb{F}_q)$, it corresponds to $(a', b') = (2/3, 1/3)$, and $E_{1/9}$ is a twist of W, as the solution $2^3 + 1^3 = 9 \cdot 1^p$ requires. Its X_1 is $-r^2/3$, which lies in $3^{-1}\mu_n(\mathbb{F}_q)$ ($3 \mid n$ when $q \equiv 1 \pmod 3$), so the refined form fails too. This is a test of the two cosets of Proposition 12.11, which must contain the ζ and the X_1 of that solution.

S17. The exponent 3, and searches. This is used in the proof of Theorem 12.1. The curve $X^3 + Y^3 = 3Z^3$ is mapped to $y^2 = x^3 - 3888$ by $x = 36Z/(X + Y)$, $y = 108(X - Y)/(X + Y)$ ($X + Y \ne 0$ unless $Z = 0$); this curve has trivial torsion and rank 0, proved by a 2-descent (PARI's `ellrank`). Also $11^3 + (-2)^3 = 3 \cdot 21^2$. Searches: over the coprime pairs with $|b| \le a \le 3000$, no $a^3 + b^3 = 3^k c^n$ with $k \ge 1$ and n divisible by a prime ≥ 5 other than $2^3 + 1^3 = 9 \cdot 1^n$; over $|x|, |y| \le 4000$ with $\gcd(x, y) = 1$, the only solutions of $x^3 + y^3 = 3^k z^n$ with $1 \le k \le 4$, $n \in \{4, 5\}$ and $z \ne 0$ are those of $9z^5$ in Theorem 12.3 (c), and $(2, 1, \pm 1)$, $(71, -23, \pm 14)$ with their images under $x \leftrightarrow y$ for $9z^4$.

S18. The exponent 4. This is used in the proof of Theorem 12.3 (a). The identity $v^2 = u^3 + 27u$ for the map from H to E was checked symbolically, and $\#E(\mathbb{F}_5) = 2$, $\#E(\mathbb{F}_7) = 8$. The homogeneous spaces $N^2 = dM^4 - (108/d)e^4$ of $\hat{E}$ have no primitive solution modulo 3^4 for $d = -1, 2, 3, -6$ and none modulo 2^5 for $d = -2, 6$ (exhaustion).

S19. The curve Γ_9. This is used in the proof of Proposition 12.13 for $j = 9$. For $L = \mathbb{Q}(12^{1/5})$: the discriminant $2^4 \cdot 3^4 \cdot 5^5$; the 29 prime ideals of norm at most 125 and an explicit generator of each; the units $\varepsilon_1, \varepsilon_2$ (norm -1, regulator $66.54\ldots$) and the seven nontrivial products of $-1, \varepsilon_1, \varepsilon_2$, none a square; π_2; the sixteen classes against the four conditions (exactly $1, -\varepsilon_1, -\varepsilon_2, \varepsilon_1\varepsilon_2$ pass); $\mathcal{I}_2$ from the $\mathbb{Q}_2$-points with $x = -199, -196$ (and, as a check, $x = -60/7, -59/5$); $\#J_9(\mathbb{F}_p)$ for $7 \le p \le 61$ from the characteristic polynomials of Frobenius, and for $p = 7, 11, 13, 41$ also from the numbers of points of Γ_9 over $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$.

S20. The curve Γ_3. This is used in the proof of Proposition 12.13 for $j = 3$. The same 2-descent as in item S19, for $L = \mathbb{Q}(6^{1/5})$, gives $\mathrm{Sel}^2(J_3/\mathbb{Q}) = 0$, and $\#J_3(\mathbb{F}_7) = 50$, $\#J_3(\mathbb{F}_{11}) = 41$.

S21. The curve Γ_1: descent and sieve. This is used in Steps 1–3 of the proof of Proposition 12.14. The 2-descent with the primes above 2, 3 and 5 (the class group of $\mathbb{Q}(18^{1/5})$ certified below its Minkowski bound 25.06; local images at 2 and 5 from points) gives a Selmer group of dimension 1; the orders $\#J(\mathbb{F}_\ell)$, $7 \le \ell \le 71$, have greatest common divisor 1. The identities of Step 2, $2D' = D$ over $\mathbb{Q}$, the order 25 of $\bar{D}'$ in $J(\mathbb{F}_7)$, the discrete logarithms at 7, and the values of τ modulo 25 at 149 were computed with an implementation of Cantor's algorithm in Python.

S22. The curve Γ_1: the 7-adic logarithms. This is used in Step 4 of the proof of Proposition 12.14. The divisors $25D$, $50D$, $175D$, $25D'$, $75D'$ were computed exactly over $\mathbb{Q}$ as Mumford pairs (u, v) (for $25D$ the numerators and denominators of the coefficients of u have about 660 bits, and those of v about 1000 bits); each is $[P_1 + P_2 - 2\infty]$ with u integral at 7, its roots $\equiv 6 \pmod 7$, and v of valuation -1 (or -2 for $175D$). The integrals $\int_{\iota P_2}^{P_1} \omega_i$ were computed by termwise integration of the expansion of $x^i/(2y)$ in $x - x(P_2)$, with integers modulo 7^{40} and 80 terms, and the branch of y was checked at P_1. This gives the digits of L_0, L_1 in Step 4, and the same method gives the consistency relations $\log(50D) = 2\log(25D)$, $\log(D) = 2\log(D')$ (through $25D'$), $\log(75D') = 3\log(25D')$ and $\log(175D) = 7\log(25D)$ modulo 7^{30}. The expansions at ∞ and at $(4, \pm 16)$ were computed to degree 44, with the valuations used by Strassmann's theorem.

S23. The curves Γ_5 and Γ_7: descent and sieve. This is used in Steps 1 and 2 of the proofs of Propositions 12.15 and 12.16. For $L = \mathbb{Q}(2^{1/5})$ and $L = \mathbb{Q}(72^{1/5})$: the discriminants; the class groups, certified below the Minkowski bounds 13.92 and 125.28 by explicit generators of the 5 and 34 prime ideals of norm below them; the units, of odd index (no nontrivial product of -1 and the two units is a square); the local factorizations of f; and the 2-descent with the primes above 2, 3 and 5, which gives Selmer groups of dimension 1, generated by the classes of $12 - \theta$ and $\theta^2 + 432$, neither a square in L. Also: $f - (72x + 1296)^2 = (x^2 + 432)(x^3 - 432x - 5184)$; $\#J_j(\mathbb{F}_\ell)$ for $7 \le \ell \le 71$, from the characteristic polynomials of Frobenius and from point counts, with greatest common divisor 1 for $j = 5$ and $j = 7$; $\bar{D}$ and $\bar{G}_1$ of order 50 in $J_j(\mathbb{F}_7)$. The $\{2, 5\}$-sieve was run with an implementation of Cantor's algorithm in Python (brute-force discrete logarithms in $J_j(\mathbb{F}_\ell)$): for Γ_5 and $7 \le \ell \le 199$ it leaves $\tau \equiv 0, \pm 1 \pmod{50}$, with cuts at $\ell = 7, 19, 59, 79, 199$; for Γ_7 and $7 \le \ell \le 149$ it leaves $\tau \equiv 0 \pmod{50}$, with cuts at $\ell = 7, 19, 43, 149$. Searches: PARI's

`hyperellratpoints` with height bound 10^5 finds on $\Gamma_1,\Gamma_3,\Gamma_5,\Gamma_7,\Gamma_9$ only $(4,\pm16)$ on Γ_1 and $(12,\pm432)$ on Γ_5, and a separate search agrees; over $|x|,|y|\le 3000$ with $\gcd(x,y)=1$, $z\neq 0$ and $0\le k\le 15$, the solutions of $x^3+y^3=3^kz^5$ are those with $k=0$ and $xy=0$ and those of Theorem 12.3 (c).

S24. The curves Γ_5 and Γ_7: the 7-adic logarithms. This is used in Step 3 of the proofs of Propositions 12.15 and 12.16. The divisors ND and NG_1 for $N=50,100,150,350$ were computed exactly over $\mathbb{Q}$ as Mumford pairs (u,v) (for $50D$ the numerators and denominators of the coefficients of u have about 650 decimal digits); each is $[P_1+P_2-2\infty]$ with u integral at 7 and $u\equiv(x-4)^2$, respectively $(x-3)^2 \pmod 7$. For Γ_5 the integrals $\int_{\iota P_2}^{P_1}\omega_i$ were computed by termwise integration of the expansion of $x^i/(2y)$ about the midpoint of $x(P_1)$ and $x(P_2)$, modulo 7^{40} with 120 terms, and the branch of y was checked at P_1. For Γ_7, x was expanded as a power series in y^2 about $(e,0)$ by series reversion, and the integrals were summed through the power sums of $y(P_1),y(P_2)$, with the check that the expansion reproduces $x(P_1)+x(P_2)$ modulo 7^{40}. The values $\log(ND)/N$, respectively $\log(NG_1)/N$, for the four N agree to the working precision, and give the digits of L_0,L_1 and the residues of α,β in the proofs. The expansions at ∞ and at $(12,\pm432)$ were computed with the valuations used by Strassmann's theorem.

The obstructing point

S25. Numerical evidence at the obstructing point. Assuming the expected functional equations of the degree-6 L-functions of $\mathrm{Ind}_K^{\mathbb{Q}}\mathbb{V}_{s_0}$ and of its twists, numerical fits with about $5\cdot10^5$ Dirichlet coefficients indicate the following: for $\mathbb{V}_{9/8}\otimes\kappa_2^{-1}$, which is unramified at 2, and for $\mathbb{V}_9$, the conductor exponent at $\mathfrak{p}$ is 14, so inertia at $\mathfrak{p}$ does not act unipotently there (the functional equation is satisfied to within about 2^{-67}). The same holds for $\mathbb{V}_{9/8}\otimes\kappa_2$, which is ramified at 2 with conductor exponent 3; its fit was confirmed with about $7\cdot10^5$ coefficients (to within about 2^{-44}, and 2^{-40} with $5\cdot10^5$). For $\mathbb{V}_{3^8}$ and $\mathbb{V}_{3^{10}}$, inside the disc, the conductor exponent is 2 and the Euler factor at $\mathfrak{p}$ has degree one with Frobenius eigenvalue 1, consistent with Theorem 4.1.

The tests of Remark 4.18 were made as follows. The Euler factors at the primes $q\le X$ not dividing $3s_0(s_0-1)$ were computed from the character sums to which Lemma 3.7 reduces, by a Gauss-sum formula checked against a direct evaluation of these sums at the small primes. At a prime $\mathfrak{q}\nmid 3$ of K dividing $s_0(s_0-1)$ the factor given by specialization was used: $1-\mathrm{N}\,\mathfrak{q}^{-s}$ if $\mathfrak{q}\mid s_0$ (Lemma 5.3 (b) and Proposition 4.16), and, if $\mathfrak{q}\mid s_0-1$, the characteristic polynomial of Frobenius on the stalk at 1 of $j_*\mathbb{V}$, computed from the same sums at $s=1$, multiplied by $1\mp\mathrm{N}\,\mathfrak{q}^{1-s}$ (both signs tried) when $3\mid v_{\mathfrak{q}}(s_0-1)$. The functional equation was tested with PARI's `lfuncheckfeq`, in 64-bit arithmetic and with both signs: for $f\le2$ with 117 candidate Euler factors at 3, which include every factor allowed by Theorem 4.1, and, at the points outside the disc, for larger f (up to 14 at $s_0=9$, 6 at -27, 4 at $-2\cdot3^7$, and $7,5,7,5,3$ at $s_0=-9,27,81,243,729$) with 46 of them; the candidates are listed in [1]. For the choices in the table of Remark 4.18, `lfuncheckfeq` returned -66 to -69, that is, the functional equation holds to about 2^{-66}; every other choice gave -17 or more, an error of at least 2^{-17}.

References

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